01 / CHARGE Type-C source gate and primary protection

File: charge_primary.kicad_sch

02 / THM6 isolated 5V supply and local decoupling

File: isolated_converter.kicad_sch

03 / Isolated secondary 2S charger and protected harness

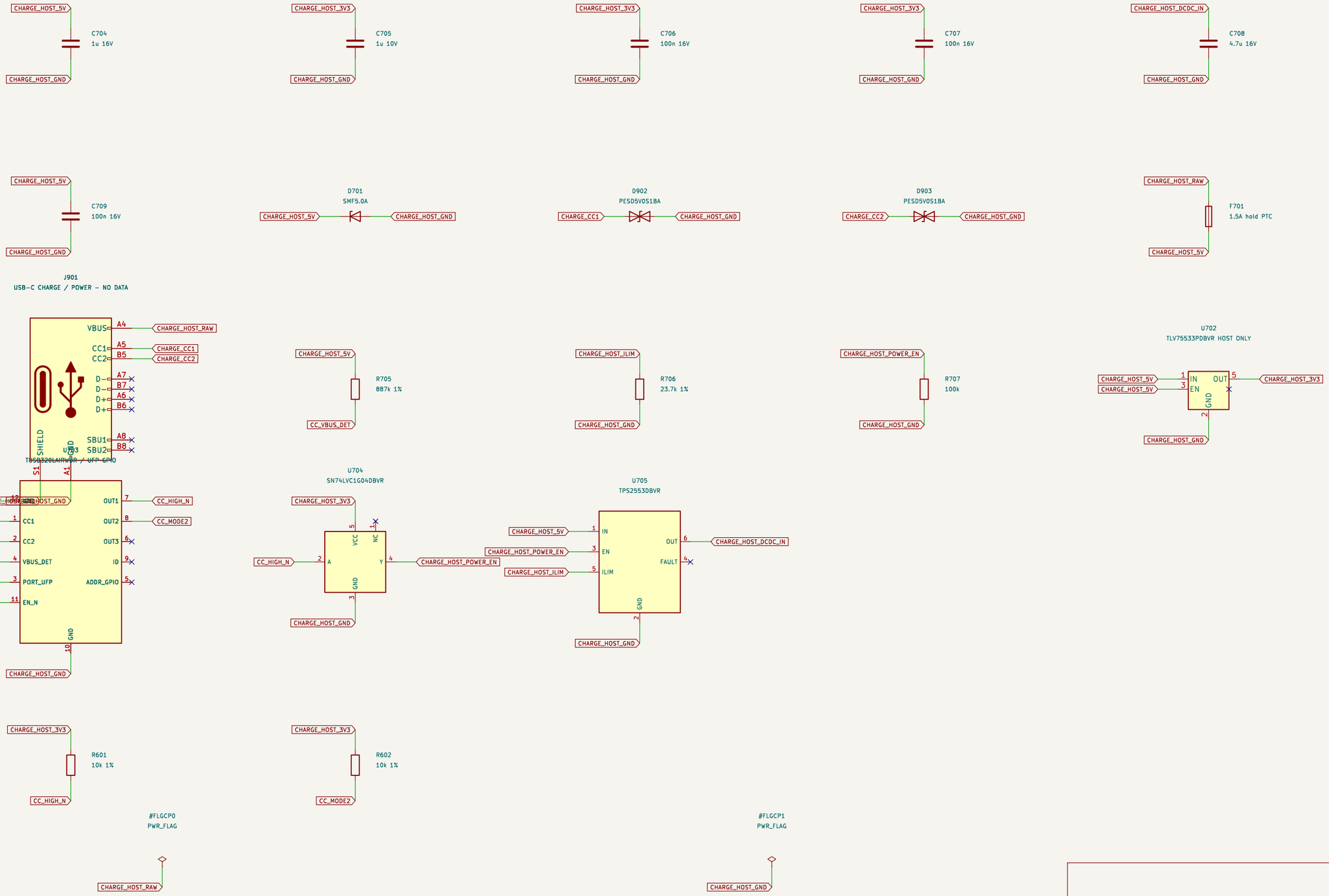
File: charger_secondary.kicad_sch

YuEEG

Sheet: /
File: YuEEG_8CH_RevF_POWER.kicad_sch

Title: Rev F detachable isolated 2S charger

Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 1/4



Primary ground is CHARGE_HOST_GND. It has NO electrical harness connection to DATA HOST_GND or system GND.
TUSB320 enables the current-limited converter input only for Type-C 1.5A/3A advertisement.

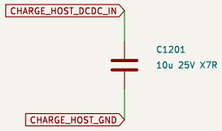
BATTERY ONLY – NOT VALIDATED FOR HUMAN CONNECTION

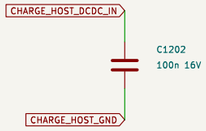
YuEEG

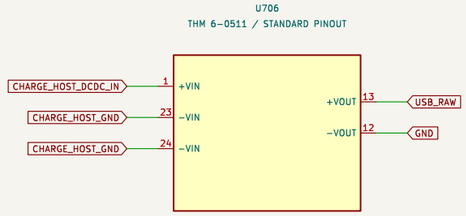
Sheet: /01 / CHARGE Type-C source gate and primary protection/
File: charge_primary.kicad_sch

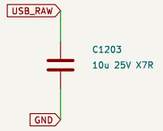
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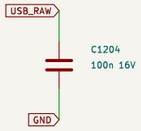
Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 2/4

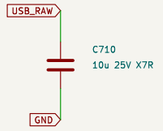






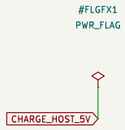












Retain 8mm clearance and creepage on every layer. No copper crosses the insulation moat.
All SMD parts TOP; the DIP module is through-hole and requires separate assembly.
6W module remains on this detachable board so its body and height do not occupy the worn acquisition board.

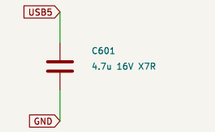
BATTERY ONLY — NOT VALIDATED FOR HUMAN CONNECTION

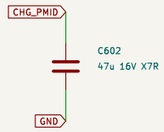
YuEEG

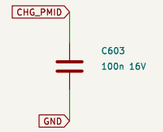
Sheet: /02 / THM6 isolated 5V supply and local decoupling/
File: isolated_converter.kicad_sch

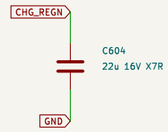
Title: 02 / THM6 isolated 5V supply and local decoupling

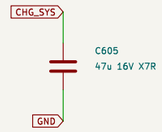
Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 3/4



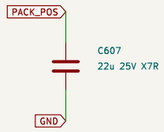


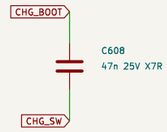


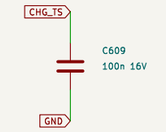




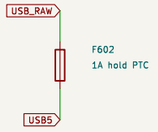


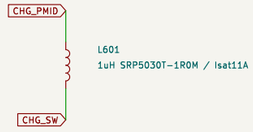




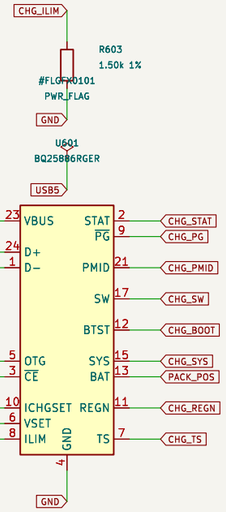


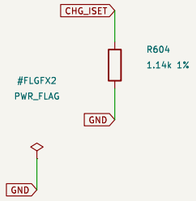




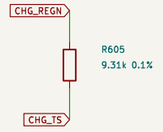


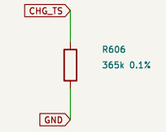


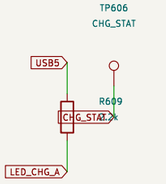


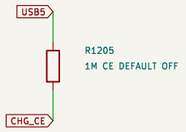


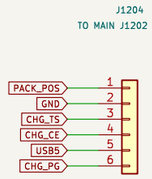












8.2V, nominal 299mA; input budget includes module loss and source gate current tolerance.
R1205 defaults CE high when the main/CE harness is disconnected. Main POWER ON inhibits charging.
SYS has only local capacitance; it cannot power the acquisition board. NTC comes from the main pack connector.
Charge only with electrodes physically disconnected. Verify GH4 pin order and NTC type before applying cell power.

BATTERY ONLY — NOT VALIDATED FOR HUMAN CONNECTION

YuEEG

Sheet: /03 / Isolated secondary 2S charger and protected harness/
File: charger_secondary.kicad_sch

Title: 03 / Isolated secondary 2S charger and protected harness

Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 4/4